Quiz #4

1. A laboratory repeatedly measures a reference mass whose exact value is known to be 50.00 g. Five measurements are

$$50.08, \quad 50.07, \quad 50.09, \quad 50.08, \quad 50.08 \text{ g}.$$

   (a) Are these measurements precise? Briefly explain.

   Yes. The measurements are **precise** because they are tightly clustered between 50.07 g and 50.09 g. Thus, the chance-error spread is small.

   (b) Do the measurements appear to be accurate? What type of measurement error is suggested by these readings?

   The measurements do not appear to be accurate. They are consistently about 0.08 g above the known value of 50.00 g. This suggests a positive **bias** (systematic error).

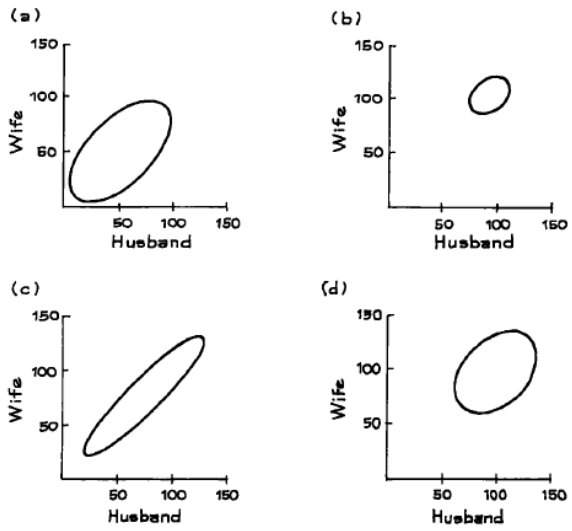
   (c) Would taking many more measurements and averaging them eliminate this problem? Briefly explain.

   No. Taking more measurements and averaging them can reduce the effect of **chance error**, but it does not remove **bias**. The instrument or measurement procedure would need to be calibrated or corrected.

2. A study of the IQs of husbands and wives obtained the following results:

$$\bar{x} = 100 \quad \text{(husbands)}, \qquad \bar{y} = 100 \quad \text{(wives)},$$

$$s_x = 15, \qquad s_y = 15, \qquad r = 0.6.$$



Which scatter diagram matches these data? Briefly explain why you reject the others.

The best match is **(d)**.

The center of the scatterplot should be near

$$(\bar{x}, \bar{y}) = (100, 100).$$

Because $s_x = s_y = 15$, the horizontal and vertical spreads should be about the same. Also, $r = 0.6$ indicates a **moderate positive linear association**, so the cloud should slope upward but should not be extremely narrow.

Diagram (a) is centered too low. Diagram (b) has too little spread. Diagram (c) shows a much stronger positive association than $r = 0.6$. Diagram (d) has the appropriate center, similar horizontal and vertical spread, and a moderate positive association.

3. Suppose `gap_2007` contains the Gapminder observations from 2007. Consider the following code:

```
result = gap_2007.loc[
    (gap_2007["continent"] == "Americas")
    & (gap_2007["pop"] > 10_000_000),
```

```
    ["country", "lifeExp", "gdpPercap"],
].sort_values("lifeExp", ascending=False)
```

(a) In words, which rows are selected?

The code selects countries that are in the `Americas` and have populations greater than 10,000,000.

(b) Which columns appear in `result`, and how are the rows ordered?

The resulting DataFrame contains the columns `country`, `lifeExp`, and `gdpPercap`. The rows are sorted by `lifeExp` from highest to lowest because `ascending=False`.

(c) Why is `&` used instead of `and` when combining the two conditions?

Each condition produces a pandas Series of Boolean values, one for each row. The operator `&` combines these Boolean Series element by element. Python's `and` is used for combining individual Boolean values, not pandas Series.